

1 Critical evaluation of the modified Gompertz response function
2 as a parametric template for treatment-response trajectories in
3 N-of-1 trial simulation

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5 August 15, 2026

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34 1 Abstract

35 **Background.** Simulation-based power analyses for aggregated N-of-1 trials require a parametric template for the within- participant response trajectory. The pmsimstats framework
36 adopts a modified Gompertz function for each of three latent response components (biological
37 response, placebo-belief response, time- variant natural-history component). The choice
38 of family is loaded into the simulated data before any analysis-model question is asked, so
39 the implications of the family for power, Type I error, and bias are foundational rather than
40 peripheral.

42 **Methods.** We evaluate the modified Gompertz family against the relevant biological, pharmacokinetic, and biostatistical literature. We characterize the parameter space the function
43 covers, the trajectories it can and cannot represent, and the sensitivity of standard pmsimstats inference to misspecification of the response-curve family. Three alternative families are
44 compared as sensitivity analyses: the logistic (symmetric sigmoidal), the hyperbolic-tangent
45 (fast-saturating), and a piecewise-linear breakpoint model (no smoothness assumption). For
46 each alternative we compute power, Type I error, and bias of the biomarker-treatment interaction estimate at trial-relevant sample sizes against simulated data drawn from the alternative
47 family but analyzed under the standard linear-mixed model.

51 **Results.** A $1\{,\}000$ -replicate-per-cell simulation across 16 cells (4 trajectory families $\times$ 2 data-generating process (DGP) architectures $\times$ 2 biomarker effect-size levels) at $N = 70$, open-label
52 plus blinded discontinuation (OL+BDC) design, $t_{1/2} = 0$, found that the inferential consequence of trajectory family is architecture-dependent. Under the covariance-moderation architecture (multivariate normal (MVN) differential correlation), where the biomarker-treatment
53 interaction lives in the covariance structure and is therefore mean-independent, trajectory
54 family is exactly inert: with matched random-number seeds the four families return identical
55 rejection decisions, so the production-seed spread $\{0.774, 0.777, 0.791, 0.804\}$ is sampling noise.
56 Under the mean-moderation architecture, implemented in its faithful multiplicative form in
57 which the biomarker moderation rides the response trajectory, the four families separate: powers were $\{0.743, 0.760, 0.778, 0.782\}$ for Gompertz, hyperbolic-tangent, piecewise-linear, and
58 logistic respectively, a 0.039 spread in which the Gompertz default sits roughly two Monte
59 Carlo standard errors (MCSE) below the logistic and is the lowest-power, that is mildly
60 conservative, family. Type I error at the null sat in $[0.019, 0.038]$ across all eight null cells,
61 slightly conservative and without family-specific inflation; bias of the interaction estimate was
62 negligible and family-invariant (≈ -0.23).

67 **Conclusions.** The modified Gompertz family is a defensible default for the response-curve
68 template in pmsimstats N-of-1 simulation, with two qualifications. Under a covariance-channel
69 (MVN) data-generating process, trajectory family cannot affect the interaction test by con-

struction, so the choice is inferentially free. Under a mean-moderation data-generating process, trajectory family does carry a modest inferential cost at the few-percent level, and the Gompertz default is mildly conservative relative to the symmetric and breakpoint alternatives. We recommend reporting Gompertz-vs-alternative sensitivity at least once per parameter regime where simulation results inform protocol-design decisions, with particular attention to mean-moderation regimes, and we propose a default sensitivity grid that the framework can run alongside its primary cells.

2 Introduction

2.1 A claim about response-curve choice in N-of-1 trial simulation

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The first claim is that the modified Gompertz function used as the parametric template for treatment-response trajectories in aggregated N-of-1 trial simulation (specifically, in the pmsimstats simulation framework that operationalizes the Hendrickson et al. design family¹) is a defensible default for the saturating-shape regime that characterizes most pharmacodynamic and chronic-condition response curves over the typical trial timescale. The function is well-established in tumor growth dynamics^{2,3}, microbial growth modeling⁴, and the broader unified-Richards growth-curve family⁵⁻⁷; its three-parameter form is flexible enough to capture the asymmetric saturation profiles that linear extrapolation, exponential decay, or symmetric logistic models cannot.

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The second claim is that family-misspecification sensitivity is essentially absent from the published N-of-1 methodology literature, and that it should be reported routinely rather than assumed away. The methodological N-of-1 corpus^{1,8-10} is, to the best of our knowledge, silent on parametric trajectory shape: papers either inherit a chosen family from the upstream literature without comparing alternatives, or specify a linear or AR(1) form that is too coarse to capture the saturation that real trajectories exhibit. The oncology-modeling literature has set the precedent for what a proper family-comparison sensitivity analysis looks like^{6,11}, and the pharmacology literature has converged on Hill / sigmoid-Emax / indirect-response models for related questions^{12,13}. The N-of-1 literature has not imported this comparative apparatus, and the present paper endeavors to supply it.

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The two claims are coupled because the case for a chosen family is incomplete without the comparison. We do not argue that Gompertz is the correct family in any strict sense; rather, we argue that it is a reasonable choice within the unified-

Richards family of saturating curves, and that the inferential consequences of choosing it (versus the symmetric logistic, the von Bertalanffy growth curve¹⁴, or a piecewise-linear breakpoint model) should be quantified rather than asserted. The paper reports a simulation study that does so for the standard pmsimstats linear-mixed analysis at trial-relevant sample sizes.

2.2 Why the question matters

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First, the chosen family loads into the data before any analysis-side question is asked. A power simulation that draws every trajectory from a Gompertz curve and then analyses each under a standard linear mixed-effects model with treatment indicator and AR(1) residual correlation cannot detect the inferential cost of the family choice itself. Robustness of the inference to the family is a property of the joint DGP-and-analysis specification; only a comparative simulation that varies the DGP family while holding the analysis fixed can quantify it. This is the same logic that underpins analysis-model misspecification studies in the broader clinical-trial methodology literature¹⁵, applied one level down in the simulation hierarchy, and it appears to us that the N-of-1 simulation literature has not yet drawn the parallel explicitly.

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Second, the function family has known identifiability fragility in short noisy trajectories. Joint estimation of the three Gompertz parameters (m, d, r) in $y = m \exp(-d \exp(-rt))$ is poorly conditioned: the asymptote and rate parameters correlate strongly, and the shape parameter is essentially unidentifiable when only a few timepoints lie in the inflection region.⁶ and⁷ propose unified-Richards reparameterizations in which the parameters become biologically interpretable (maximum relative growth rate, age at inflection, value at inflection, asymptote) with lower correlation, and¹⁶ develops the formal ‘sloppiness’ diagnostic that quantifies this directly as a property of the parameter space.¹¹ fitted eight tumor-growth models (exponential, exponential-linear, power-law, Gompertz, logistic, generalized logistic, von Bertalanffy, dynamic-carrying-capacity) to two in vivo systems and ranked them on goodness-of-fit, parametric identifiability, and predictive power; the power-law model was the most identifiable in lung data while Gompertz was best for breast. The same kind of comparative diagnosis has not been performed for short N-of-1 trajectories, and its absence is, in our view, problematic.

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Third, the default in N-of-1 simulation reflects historical borrowing rather than empirical optimality. The Gompertz form entered the pmsimstats framework via

the broader growth-curve tradition, where its use traces to Laird's tumor dynamics work² and to Zwietering's microbial growth reformulations⁴. The pharmacology literature, addressing related questions about dose-response and time-course of pharmacological action, defaults to the Hill / sigmoid-Emax family^{12,17} and to Jusko's indirect-response models¹³. These two traditions have largely not spoken to each other; the N-of-1 simulation literature has inherited the growth-biology default without an explicit defense against the pharmacology default. Either default may be adequate for the saturation regime, but the inheritance is not itself evidence, and the question merits a direct empirical answer.

2.3 What the published N-of-1 literature does and does not do

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A review of the recent N-of-1 methodology literature confirms, to the best of our knowledge, a silence on the parametric-trajectory-family question.

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The CONSORT extension for N-of-1 trials¹⁰ codifies reporting standards but does not address parametric trajectory shape. The foundational programmatic statements^{8,9} emphasize design and analysis broadly without committing to a parametric response form. The Hendrickson et al. framework¹ introduces the modified Gompertz template into the simulation engine for biomarker-validation power analyses without comparing it against alternatives. Recent applied papers in chronic-condition N-of-1 trials (including the prazosin-PTSD reference work building on¹⁸) inherit the framework without examining the choice. Unfortunately, this pattern of silent inheritance is pervasive enough that no published N-of-1 simulation study, to the best of our knowledge, has broached the family-misspecification question.

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In adjacent literatures the comparative apparatus does exist. The oncology-modeling community routinely fits multiple growth families and compares them on identifiability and predictive performance¹¹. The pharmacometrics community produces explicit sensitivity studies for sigmoid-Emax and indirect-response models^{19,20}. The latent-class growth-trajectory literature treats response-curve heterogeneity through mixture modeling²¹. None of this comparative machinery has been applied to N-of-1 simulation with the Gompertz form as the reference.

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The present paper's contribution is to import that machinery, specifically: a four-family comparison (Gompertz, symmetric logistic, hyperbolic-tangent, piecewise-linear breakpoint) crossed against the standard pmsimstats linear-mixed analysis,

189 with results reported in the Morris-White-Crowther¹⁵ ADEMP framework already
190 adopted as the project-wide simulation-reporting standard.

191 2.4 What this paper contributes

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194 In section 2 we characterize the modified Gompertz family as implemented in
195 pmsimstats: the vertical-offset adjustment, the parameter regions corresponding
196 to the calibration sets used across published pmsimstats simulation studies, and
197 the unified-Richards reparameterization^{6,7} that supplies more interpretable and
198 less correlated parameters.

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201 In section 3 we survey three alternative response-curve families: the symmetric
202 logistic²², the hyperbolic-tangent (formally a rescaled symmetric logistic), and a
203 piecewise-linear breakpoint model. We present each at matched marginal effect
204 size against the Gompertz reference, with the shapes overlaid for visual compari-
205 son.

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208 In sections 4 and 5 we conduct a Monte Carlo sensitivity-analysis simulation that
209 generates trajectories under each of the four families and analyses them under the
210 standard pmsimstats linear-mixed model. We report power, Type I error, and bias
211 of the biomarker-treatment interaction estimate under each cell, with conventional
212 MCSEs. The simulation borrows the comparative methodology of¹¹ (eight-model
213 tumor-growth comparison) and the sloppiness diagnostic of¹⁶ for small-sample
214 identifiability assessment.

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217 In section 6 we identify the trajectory regimes in which Gompertz misspecifica-
218 tion would be expected to be consequential: non-saturating growth beyond the
219 trial window, biphasic patterns, and breakpoint behavior. For these regimes, the
220 standard N-of-1 inference is materially affected by the family choice, and we recom-
221 mend specific sensitivity-grid extensions that pmsimstats users should run along-
222 side their primary cells.

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225 The full mathematical and historical context, including citations that did not
226 survive editing into the manuscript proper, is preserved in the companion technical
227 document at docs/25-gompertz-model-evaluation.md.

3 The modified Gompertz function in pmsimstats

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The Gompertz curve originates in Gompertz’s 1825 law of human mortality²³ and was first applied as a general growth curve by²⁴; the pmsimstats framework adopts the modified, monotone-saturating form of that curve. The framework parameterizes each of the three latent response components (biological response BR, placebo-belief response PB, and time-variant natural-history component TV) with a modified Gompertz function

$$y(t) = \text{maxr} \cdot \exp(-\text{disp} \cdot \exp(-\text{rate} \cdot t)), \quad (1)$$

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with vertical-offset adjustment so that $y(0) = 0$. The three parameters are maxr (asymptotic maximum response, matching the on-drug or on-natural-history saturation level), rate (which controls how quickly the response approaches its asymptote), and disp (which controls the curvature near $t = 0$). The Gompertz form is monotone-saturating and asymmetric: the inflection occurs at $t^* = \log(\text{disp})/\text{rate}$, with the rise being relatively slower before the inflection and faster after. The parameterization used at the prazosin/PTSD reference cell sets $\text{maxr}_{BR} \approx 4$ nightmares per week, $\text{rate}_{BR} \approx 0.5$ per week, and $\text{disp}_{BR} \approx 4$ for biological response; analogous parameter sets for PB and TV are documented in docs/25-gompertz-model-evaluation.md and R/utilities.R.

4 Alternative response-curve families

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Three alternative families serve as sensitivity-analysis comparators, and in this section we shall describe each in turn before the simulation study proper. Each is parameterized so that the asymptotic maximum and a single-anchor calibration point (the response level at $t = 5$ weeks, calibrated to the Gompertz family at the same anchor) are matched across families. That is, the families differ in shape envelope (smoothness, symmetry around the inflection, and curvature) while sharing the same marginal effect size at the anchor, a design choice that we discuss further in section 4.

4.1 Symmetric logistic

The logistic function

$$y(t) = \frac{\text{maxr}}{1 + \exp(-\text{rate} \cdot (t - t_0))} - \text{offset} \quad (2)$$

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is monotone-saturating and symmetric around its inflection at $t = t_0$. A vertical offset is subtracted so that $y(0) = 0$. At matched maxr and matched response level at the calibration anchor, the logistic family rises faster than Gompertz before the inflection and slower after, producing a visibly different shape envelope while sharing the same asymptote. The symmetric inflection is the principal qualitative contrast with the asymmetric Gompertz curve, and the reader will note that this symmetry property is, in a sense, the logistic family's most theoretically interesting feature for the purposes of the present comparison.

4.2 Hyperbolic-tangent

The hyperbolic-tangent variant

$$y(t) = \text{maxr} \cdot \frac{\tanh(\text{rate} \cdot (t - t_0)) + 1}{2} - \text{offset} \quad (3)$$

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is also monotone-saturating and symmetric, but with a qualitatively faster approach to the asymptote than the logistic. The $\tanh$ family produces the steepest pre-asymptote rise of the three smooth families, and it has been used in pharmacokinetic modeling for drugs with rapid receptor saturation. We include it here as the representative of the fast-saturating smooth regime that neither the logistic nor the Gompertz family can span with their standard parameterizations.

4.3 Piecewise-linear breakpoint

The piecewise-linear-breakpoint family, formalized in the broken-line regression literature²⁵,

$$y(t) = \text{maxr} \cdot \min(1, t/t_{\text{bp}}) + \text{post_slope} \cdot \max(0, t - t_{\text{bp}}) \quad (4)$$

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is the canonical non-smooth alternative: a linear ramp up to a breakpoint t_{bp} , then constant (when $\text{post_slope} = 0$) or with a small post-breakpoint slope. The breakpoint is the only non-differentiable feature across the four families and is the principal qualitative contrast with the three smooth families. This family is the strongest available test of the framework's sensitivity to non-smoothness in the trajectory. But how different is the piecewise-linear breakpoint from the smooth

296 families, from the perspective of the Wald test that the analysis model applies?
297 That question is precisely what the simulation in section 4 is designed to answer.

298 5 Sensitivity-analysis simulation design

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301 The primary estimand is the biomarker-treatment interaction coefficient $\hat{\beta}_{bm:D}$
302 from the linear mixed-effects analysis model $Sx \sim bm + t + Dbc + bm:Dbc +$
303 $(1|ptID)$ with `corCAR1(form = ~t|ptID)`. Primary inferential outputs are power
304 for $H_0 : \beta_{bm:D} = 0$ at $\alpha = 0.05$, and Type I error under $\beta_{bm:D} = 0$. Secondary
305 estimands are bias of $\hat{\beta}_{bm:D}$ relative to the calibrated true effect and the empir-
306 ical MCSE of $\hat{\beta}_{bm:D}$ across replicates. Performance measures are reported with
307 binomial-proportion MCSEs for proportions and the standard MCSE of the within-
308 replicate mean for continuous quantities.

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311 The full Aims, Data-generating mechanisms, Estimands, Methods, and Perfor-
312 mance measures are pre-registered at `analysis/scripts/gompertz-evaluation/00-ademp-pre-registr`
313 which fixes the four DGP families (modified Gompertz, symmetric logistic,
314 hyperbolic-tangent, piecewise-linear breakpoint), the calibration sub-study
315 (Study 0) that matches all four to a common marginal effect size, the four-family
316 power-and-bias factorial (Study 1, 144 cells), and the sloppiness eigenvalue
317 diagnostic (Study 2) before the production runs.

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320 For the present submission the simulation is scoped to a 16-cell focused factorial:
321 four trajectory families crossed with two DGP architectures (mean modera-
322 tion and MVN differential correlation, following the companion manuscript
323 at `analysis/report/01-dgp-mean-moderation-vs-mvn/`) crossed with two
324 biomarker effect-size levels ($c_{bm} \in \{0, 0.45\}$). The trial design is the prazosin-
325 calibrated Hybrid OL+BDC, $N = 70$, $t_{1/2} = 0$, with carryover set to absent in
326 order to isolate the trajectory-shape effect from carryover-mediated correlation
327 decay. Following the compendium convention, N is the total across randomization
328 paths: the driver allocates 35 participants to each of the design's two paths, and
329 the paths are concatenated before analysis. Each cell is run at $n_{\text{reps}} = 1,000$
330 replicates with 8-core `parallel::mclapply`. The full 144-cell grid (sample sizes,
331 effect sizes, DGP-side carryover) anticipated in the original abstract is deferred
332 to a follow-up; the focused 16-cell factorial answers the central scientific question
333 (does trajectory family choice matter for the `bm:Dbc` Wald test?) at MCSE
334 0.013–0.016 per cell.

Notation for the moderation parameter. Because this factorial crosses both architectures, c_{bm} is used here as a common label for the biomarker-moderation strength at a matched numeric value. Strictly, the parameter is the correlation c_{bm} under the covariance-moderation architecture and the dimensionless multiplier β_{bm} under the mean-moderation architecture, the two being calibrated to a common scale because the Architecture-A shift is applied as $\beta_{bm} b_i \sigma_{BR}$ on the standardized biomarker b_i . The companion architecture manuscript keeps the symbols distinct; we adopt the shared label only where a statement applies to both architectures at the matched value, and the architecture is named explicitly wherever a result is architecture-specific.

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Calibration. The four trajectory families are matched at a single anchor point ($t = 5$ weeks): each family's response level at $t = 5$ is set equal to the Gompertz reference level at $t = 5$ via uniroot (logistic, hyperbolic-tangent) or closed form (piecewise-linear breakpoint). The asymptotic maximum is held constant across families. Calibrated parameters are recorded in the driver analysis/scripts/gompertz-evaluation/02-faithful-trajectory-sweep.R.

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Trajectory-family encoding. The family is routed through the data-generating process rather than imposed post hoc: it sets the BR mean curve inside the covariance construction via the `br_family` argument to `buildSigma`. Under the mean-moderation architecture the biomarker-treatment interaction is implemented in the multiplicative form of the companion manuscript at `analysis/report/01-dgp-mean-moderation-vs-mvn/`, $Y_{it} = \mu_0 + \beta_1 D_{it}(1 + \beta_{bm} B_i) + \epsilon_{it}$, so the biomarker-dependent component of the treatment effect rides the response trajectory $BR(t)$ rather than entering as a flat shift. We normalize the per-timepoint moderation so that its mean on-drug magnitude equals σ_{BR} , holding the marginal interaction effect size fixed across families while letting the temporal profile vary by family shape. Under the covariance-moderation architecture (MVN differential correlation) the interaction is encoded in the BM-BR covariance block, which is independent of the BR mean; the family therefore enters only through the mean curve and, as we verify by a matched-seed check, cannot move the interaction test.

6 Results

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The simulation program was executed at 1{,}000 replicates per cell across the 16-cell trajectory-family-by-architecture-by- effect-size factorial in 597 s using

8-core parallel::mclapply. Convergence was 100% across all 16{,}000 fits. Per-replicate output is at analysis/data/quick-sim/07-gompertz-faithful-replicates.rds; the Morris ADEMP cell-level summary is at analysis/data/quick-sim/07-gompertz-faithful-summary.rds. Cell MCSE on power runs from 0.013 to 0.014 across the alternative cells and from 0.004 to 0.006 on the conservative null cells. We begin the results section with power, then turn to Type I error, and finally consider bias of the interaction estimate.

6.1 Power across response-curve families

Trajectory-family power at $c_{bm} = 0.45$, percent $\pm$ MCSE. The mean-moderation architecture is the faithful trajectory-scaled mean-moderation form; the covariance-moderation architecture is the MVN covariance channel:

Family	Mean	Covariance
Gompertz	74.3 ± 1.4	77.4 ± 1.3
Logistic	78.2 ± 1.3	79.1 ± 1.3
Hyperbolic tangent	76.0 ± 1.4	77.7 ± 1.3
Piecewise-linear breakpoint	77.8 ± 1.3	80.4 ± 1.3

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The two architectures behave qualitatively differently. Under the covariance-moderation architecture (MVN differential correlation), the biomarker-treatment interaction is carried entirely by the covariance structure, which is independent of the BR mean curve; consequently the trajectory family, which sets that mean curve, cannot alter the interaction test. We confirm this directly: in a matched-seed check the four families return bit-for-bit identical rejection decisions, so the production-seed cell spread (0.774 to 0.804) is sampling noise rather than a family effect. Under the mean-moderation architecture, implemented in the faithful multiplicative form in which the biomarker-dependent component of the treatment effect rides the response trajectory, the families separate: power ranges from 0.743 (Gompertz) through 0.760 (hyperbolic-tangent) and 0.778 (piecewise-linear) to 0.782 (logistic), a 0.039 spread. The Gompertz-logistic gap of 0.039 is approximately two standard errors of the difference ($SE \approx 0.019$), and the Gompertz default is the lowest-power, that is the mildly conservative, family of the four. The pre-registered hypothesis that the piecewise-linear breakpoint family would behave qualitatively differently from the smooth families is not supported in the direction anticipated: the breakpoint family tracks the logistic and the hyperbolic-tangent, not the Gompertz default.

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An earlier draft of this manuscript reported essentially no family effect under either architecture. That result is now understood to have been an artifact of the interaction encoding rather than a property of trajectory family. In that draft the Architecture-A biomarker moderation was applied as a flat per-timepoint shift of magnitude $\beta_{bm}\sigma_{BR}$, independent of the response trajectory, so that the trajectory family, which enters only through the BR mean curve, could not affect the `bm:Dbc` interaction under the mean-moderation architecture any more than under the mean-independent the covariance-moderation architecture. Implementing the mean-moderation architecture in its faithful multiplicative form (section 4), in which the biomarker-dependent component scales with the response trajectory in keeping with the companion manuscript at `analysis/report/01-dgp-mean-moderation-vs-mvn/`, the mean-moderation families separate at the few-percent level while the covariance-channel invariance under the covariance-moderation architecture is preserved, the latter being a genuine mean-independence property rather than an artifact.

6.2 Type I error across response-curve families

Type I error at $c_{bm} = 0$, percent $\pm$ MCSE:

Family	Mean	Covariance
Gompertz	1.9 ± 0.4	2.6 ± 0.5
Logistic	3.8 ± 0.6	2.2 ± 0.5
Hyperbolic tangent	2.7 ± 0.5	2.6 ± 0.5
Piecewise-linear breakpoint	2.8 ± 0.5	2.7 ± 0.5

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All eight null cells produce empirical Type I error in $[0.019, 0.038]$, slightly conservative relative to the nominal 0.05 and consistent with the `corCAR1` small-sample degree-of-freedom accounting at the prazosin-calibrated $N = 70$ and 8-timepoint short serial series. At the null the biomarker moderation term is zero, so the trajectory-scaling that distinguishes the Architecture-A families under the alternative is inactive here, and the null cells differ across families only by sampling noise. There is no family-specific inflation; the highest cell, logistic under the mean-moderation architecture at 0.038, is within two MCSE of the others. The conservatism is shared across all four families, supporting the interpretation that it reflects the `corCAR1` d.f. mechanism rather than a trajectory-shape artifact, and it has appeared consistently across `pmsimstats` simulation runs at the reference parameters.

6.3 Bias of the biomarker-treatment interaction estimate

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442 Across the eight alternative cells the mean estimated $\hat{\beta}_{bm:D_{bc}}$ at $c_{bm} = 0.45$ varies
443 over a narrow range (roughly -0.230 to -0.239), a cell-to-cell spread of about 0.009
444 that is itself within the Monte Carlo noise of the estimator and shows no family-
445 specific drift; the Gompertz and logistic cells that differ most in power have nearly
446 identical mean estimates. The empirical SE of $\hat{\beta}_{bm:D_{bc}}$ is essentially constant across
447 families (SD in the range 0.073 to 0.083 across cells) and roughly equal between
448 architectures. That the Architecture-A power separation appears with negligible
449 bias and near-constant empirical SE indicates it is a precision effect operating
450 through the test statistic's sensitivity to the temporal profile of the signal, not a
451 shift in the estimand or its sampling dispersion. The matched-anchor calibration
452 holds the marginal effect size fixed across families by construction, so the residual
453 family differences are differences in the temporal shape of the moderation, which
454 is precisely the axis the design isolates.

455 7 Discussion

456 7.1 An architecture-dependent finding

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459 The central finding is that the inferential consequence of trajectory-family choice
460 is architecture-dependent. At the prazosin-calibrated reference cell (Hybrid
461 OL+BDC, $N = 70$, $c_{bm} = 0.45$, $t_{1/2} = 0$, matched-anchor calibration), the four
462 families span the qualitative range of plausible templates for monotone-saturating
463 treatment-response trajectories: asymmetric (Gompertz), symmetric smooth
464 (logistic and hyperbolic-tangent), and non-smooth (piecewise-linear breakpoint).
465 Under the covariance-moderation architecture, where the interaction is carried by
466 the BM-BR covariance and is therefore independent of the BR mean curve, the
467 family is inert by construction. Under the mean-moderation architecture, where
468 the interaction is a mean effect that, in its faithful multiplicative form, scales
469 with the response trajectory, the families separate over a 0.039 power range and
470 the Gompertz default is the lowest-power, that is the mildly conservative, family.
471 We would not wish to overstate the Architecture-A effect; it is modest, on the
472 order of two Monte Carlo standard errors at this sample size and effect size, and
473 the conditions under which it would grow or shrink are discussed in section 6.3.
474 But it is real and direction-consistent, and it overturns the apparent null reported
475 in the earlier flat- moderation draft.

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478 The pre-registered hypothesis that the piecewise-linear breakpoint family, with its
479 non-differentiable feature, would behave qualitatively differently from the smooth

families is not supported in the direction anticipated. The breakpoint family carries no distinctive power signature attributable to its non-smoothness; under the mean-moderation architecture it sits with the logistic and hyperbolic-tangent families, above the Gompertz default, and under the covariance-moderation architecture it is inert like the others. The operative axis is therefore not smoothness but the temporal profile of the biomarker-moderated signal: families whose response rises more gradually across the on-drug timepoints, and so carry more of the moderated signal at the later, more informative timepoints, yield marginally higher power than the front-loaded Gompertz curve, which reaches near-asymptote early.

7.2 Why the architectures differ

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We see two mechanisms that, taken together, account for the architecture contrast, and we shall work through each in turn. The first is the exact mean-independence of the covariance channel under the covariance-moderation architecture; the second is the residual profile-sensitivity of the trajectory-scaled mean channel under the mean-moderation architecture, which the matched-anchor calibration and the time-averaging of the Wald test damp but do not eliminate.

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The first mechanism damps the Architecture-A effect: calibration removes the amplitude differences, leaving only profile differences. The four families are matched at the asymptotic maximum and at a single anchor point ($t = 5$ weeks), and the trajectory-scaled moderation is normalized to a common mean on-drug magnitude. At matched marginal effect size, the residual within-trajectory variation in shape is small relative to the between-replicate variation in the `bm:Dbc` estimator. The calibration constraint is intentional in the simulation design: it isolates trajectory-shape effects from amplitude effects, so that the family-specific power differences we observe under the mean-moderation architecture can be attributed to the temporal profile of the moderation alone, and explains why that effect is on the order of a few percentage points rather than larger.

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The second mechanism, also damping, is that the `bm:Dbc` Wald test averages over time. The continuous D_{bc} predictor multiplied by the biomarker integrates the trajectory's contribution across all on-drug timepoints, with weights determined by the predictor's exponential-decay structure. Trajectory-shape differences (where one family rises faster than another between timepoints) are partly averaged within the integration, so the Wald test is relatively, though not perfectly, insensitive to

520 profile features. Under the mean-moderation architecture the averaging is incom-
521 plete: the residual profile difference survives into the test statistic and produces
522 the observed few-percent power spread. Under the covariance-moderation archi-
523 tecture there is no profile term to survive, because the interaction does not enter
524 the mean at all.

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527 Mean-independence applies specifically to the covariance-moderation archi-
528 tecture: the covariance moderation is upstream of trajectory shape. Under
529 MVN differential correlation, the biomarker-treatment information lives in the
530 $\partial\text{Cor}(BM, BR)/\partial D_{bc}$ structure, which is independent of the trajectory family
531 that BR follows; the family enters only the BR mean, to which the interaction
532 test is invariant. This is not a small-effect statement but an exact one, and
533 we verify it directly: holding the random-number seed fixed across families, the
534 four Architecture-B cells return bit-for-bit identical rejection decisions. The
535 production-seed spread under the covariance-moderation architecture is therefore
536 sampling noise, whereas the Architecture-A spread is a genuine, if modest, profile
537 effect.

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540 The combined effect is that trajectory-family choice is inferentially free under
541 the covariance channel and carries a modest, mildly conservative, cost under the
542 mean channel. The framework's modified Gompertz default is robust by exact
543 construction under the covariance-moderation architecture, and under the mean-
544 moderation architecture it is the lowest-power of the four families, which makes
545 it a safe rather than an optimal default: a trial designer who adopts it under
546 a mean-moderation DGP will, if anything, slightly under-state achievable power.
547 We turn next to the conditions under which the Architecture-A effect would be
548 expected to grow.

549 7.3 Conditions that would change the answer

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552 Four conditions, none exercised in the present 16-cell factorial, would plausibly am-
553 plify the modest Architecture-A family effect, and each merits a brief comment.
554 Because the covariance-moderation architecture is mean-independent, the first
555 three conditions (sample size, effect size, carryover) bear on the mean-moderation
556 architecture only; the fourth, non-monotone shape, is the only one that could in
557 principle reach the covariance-moderation architecture, and then only by changing
558 the covariance construction rather than the mean.

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561 First, smaller sample size. At $N = 70$ the `bm:Dbc` test integrates over a moderately
562 large within-subject sample; smaller N might amplify the local-shape sensitivity
563 by concentrating the temporal information at fewer timepoints where the families
564 differ more visibly.

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567 Second, larger or smaller c_{bm} . The $c_{bm} = 0.45$ reference cell sits in a power regime
568 where families have comparable signal-to-noise. Pushing c_{bm} to 0.30 (where power
569 lies near 0.5) might surface family differences via different finite-sample sampling
570 distributions, and it is possible that the tight clustering we observe is a feature of
571 this particular region of the effect-size space.

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574 Third, carryover-induced correlation decay. With $t_{1/2} > 0$, the `bm:Dbc`-mediated
575 information channel is eroded differently for different trajectory shapes, because
576 the integration weights interact with the off-drug exponential decay. The $t_{1/2} = 0$
577 slice exercised here suppresses this interaction entirely.

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580 Fourth, non-monotone or biphasic shape envelopes. The four families we have
581 examined share a monotone-saturating envelope. Families that peak and decay
582 (biphasic) or oscillate would produce genuinely different trajectories with different
583 `bm:Dbc` information content, and the Wald test would almost certainly respond to
584 those differences.

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587 Each condition is the natural axis of a follow-up sensitivity analysis. The 144-
588 cell grid anticipated in the abstract ($\$N \times c_{\{bm\}} \times t_{\{1/2\}} \times \$$ family) is the
589 canonical follow-up, and we recommend it as the production-grade extension that
590 would close out the trajectory-family question at a level of rigor adequate for the
591 methodological literature.

592 7.4 The prazosin/PTSD reference cell

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595 For the motivating prazosin-PTSD application that drives the `pmsimstats` research
596 program, the present result implies that sensitivity to the modified Gompertz
597 assumption is small relative to sensitivity to the carryover specification (the subject
598 of the companion manuscript at `analysis/report/02-carryover-sensitivity/`)

599 and to the DGP architecture (the subject of `analysis/report/01-dgp-mean-moderation-vs-mvn/`).
600 Of the three principal axes of sensitivity uncertainty in the pmsimstats simulation
601 framework, trajectory family choice is, at the reference parameters, the least
602 consequential by a substantial margin. It is not, however, entirely free: under the
603 mean-moderation DGP the modified Gompertz default is the lowest-power family
604 by a few percentage points, so a trial designer who adopts it is making a mildly
605 conservative choice. Under the covariance-channel DGP the choice is inferentially
606 free. We would suggest that sensitivity-analysis attention be directed chiefly to
607 the carryover and DGP-architecture axes, where the inferential consequences are
608 larger, while reporting at least one Gompertz-vs-alternative comparison whenever
609 the operative DGP is mean moderation.

610 8 Conclusions

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613 First, the modified Gompertz function is a defensible parametric template for
614 monotone-saturating treatment-response trajectories in N-of-1 simulation, with an
615 architecture-dependent qualification. At the prazosin/PTSD reference cell with
616 matched-anchor calibration, the four-family comparison (Gompertz, symmetric
617 logistic, hyperbolic-tangent, piecewise-linear breakpoint) produces exactly equiva-
618 lent inference under the covariance-channel architecture, where the interaction is
619 mean-independent, and a modest 0.039 power spread under the mean-moderation
620 architecture, where the biomarker moderation rides the response trajectory. In the
621 latter case the Gompertz default is the lowest-power of the four families, which
622 makes it a mildly conservative rather than an optimal default. The pre-registered
623 hypothesis that the piecewise-linear breakpoint family would behave qualitatively
624 differently from the smooth families is not supported in the direction anticipated.

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627 Second, the finding is methodologically informative precisely because it is
628 architecture-dependent. It implies that the `bm:Dbc` Wald test is exactly robust to
629 trajectory-family choice when the interaction is encoded in the covariance, and
630 only mildly sensitive to it, at the level of a few percentage points of power, when
631 the interaction is a trajectory-scaled mean effect. We cannot yet establish at
632 what sample sizes, effect sizes, or carryover levels the Architecture-A effect would
633 grow; that question is deferred to the 144-cell follow-up. Trial designers using the
634 pmsimstats framework can adopt the modified Gompertz default with confidence
635 under a covariance-channel DGP, and with the understanding that it is mildly
636 conservative under a mean-moderation DGP, while directing sensitivity-analysis
637 attention to the carryover and DGP-architecture axes that have larger inferential
638 consequences.

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641 Third, the 144-cell extended grid (sample sizes, effect sizes, DGP-side carryover)
642 anticipated in the original abstract is the natural follow-up, and we regard it as the
643 work that would be needed to close the trajectory-family question at a level of rigor
644 adequate for the methodological literature. The present focused 16-cell factorial
645 addresses the central scientific question with statistical clarity; the extended grid
646 would map the boundary conditions at which trajectory-family choice begins to
647 matter.

648 9 Conflict of interest

649 The authors declare no conflicts of interest.

650 10 Funding

651 This work received no specific funding.

652 11 Data availability statement

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655 The data and code that support the findings of this study are openly avail-
656 able in the pmsimstats-ng project repository. Per-replicate Monte Carlo outputs
657 and ADEMP-compliant cell-level summaries are versioned under `analysis/data/`;
658 simulation drivers are at `analysis/scripts/`. Exact paths for this manuscript
659 are listed in the Reproducibility section below.

660 12 Reproducibility

661 *Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt*
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663 This manuscript was rendered with R 4.4.x inside the project’s zzzcollab Docker con-
664 tainer (image hash recorded in `Dockerfile`; package set captured in `renv.lock`).
665 Principal packages used by the simulation pipeline are `nlme` (continuous-time
666 linear-mixed analysis with `corCAR1` residual correlation), `parallel` (8-core
667 `mclapply` for parameter-grid sweeps), `data.table` (the `original-extended`
668 simulation collection), `furrr` and `purrr` (the `tidyverse` simulation collection),
669 and `rmarkdown` plus `bookdown` for the manuscript build.

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672 The simulation driver analysis/scripts/gompertz-evaluation/02-faithful-trajectory-sweep.R
673 sets `set.seed(20260617)` and assigns each of the 16 cells a distinct seed off-
674 set, so replicate streams are independent across cells but reproducible. The
675 trajectory family is routed through the data-generating process via the
676 `br_family` argument to `buildSigma`, and the Architecture-A moderation is
677 trajectory-scaled via the `moderation_scaling = "trajectory"` argument to
678 `generateData` (both default to the original behavior, so all other `pmsimstats`
679 runs are unchanged). Per-replicate simulation outputs are checkpointed at
680 `analysis/data/quick-sim/07-gompertz-faithful-replicates.rds`; Morris
681 ADEMP cell-level summaries are at the companion `07-gompertz-faithful-summary.txt`.
682 The pre-registered ADEMP plan is at `analysis/scripts/gompertz-evaluation/00-ademp-pre-registr`
683 and deviations from it are recorded in `02-deviations-log.md`. The manuscript
684 source, paper-specific bibliography (`references.bib`), and SIM CSL file are at
685 `analysis/report/07-gompertz-evaluation/`.

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